

Iago Suárez

STAFF MACHINE LEARNING ENGINEER

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Summary

Staff Machine Learning (ML) Engineer with 9+ years of experience in Computer Vision, Deep Learning, and SLAM. Proven track record of leading teams, driving impactful research with publications in top-tier venues such as ICCV or IROS, and shipping products to millions of users.

Skills: PyTorch, Transformers, NeRFs, Weights & Biases, TensorFlow, Python, Modern C++, OpenCV, Eigen, Git, CMake.

Experience

Qualcomm XR Labs Europe

STAFF ML ENGINEER (PROMOTED FROM SENIOR ML ENG. IN 2024)

Madrid, Spain

Apr. 2023 – Present

- Architected SLAM feature descriptors using **PyTorch**, achieving state-of-the-art accuracy with 5 ms latency.
- Drove on-device optimization strategy across multiple neural components, applying hashing, quantization, and novel layers (pending patent).
- Led Android XR efforts for low-power **3D reconstruction** and **semantic understanding**, integrating ML pipelines into the OpenXR services.
- Contributed to a **3D-LM Transformer** for room-layout estimation inspired by SceneScript, re-architected for efficient on-device inference.

Machine Learning Circle

PRESIDENT (NON-PROFIT)

Madrid, Spain

Sep. 2022 – Present

- Created a typeface style encoder–decoder using VAEs, GANs. Tools: **Diffusers, Weights & Biases**.
- Researched vector-space typeface generation using LLMs with **LoRA, diffusion transformers**, and **differentiable rasterization**, achieving improved curve continuity and alignment over SOTA. Work under publication review.
- Main organizer of the Madrid ML Meetup, a venue where we regularly review papers around **NeRF, Gaussian Splatting**, LLMs, and Gen. AI.

New Horizon Technologies

COMPUTER VISION TEAM LEAD

Madrid, Spain

Jan. 2022 – Apr. 2023

- Managed a **7-engineer team** delivering video-surveillance and medical-imaging products end-to-end.
- Created a person-tracking system for omnidirectional cameras on **TensorRT** with custom **YOLOv5** and **DeepSORT**.
- Built a large-scale 3D heart reconstruction pipeline that has analyzed 1.3M+ echocardiography frames across four hospital groups using **U-Nets, RANSAC**, and a **3D morphable model**. (video)
- Prepared assets and technical due diligence leading to multimillion-euro acquisition by an Emirati company.

The Grafter / Bookker

COMPUTER VISION ENGINEER

Madrid, Spain

Sep. 2016 – Dec. 2021

- Developed local feature detectors and descriptors with SOTA accuracy–efficiency trade-offs; contributed to OpenCV.
- OCR & Pattern Recognition algorithms for Android using Boost, JNI, and Android Studio, deployed to 10K+ users.
- Developed customized **OpenGL/C++** visual effects using GLSL and GLESv2.

XOIA Software Development

CO-FOUNDER

A Coruña, Spain

Sep. 2015 – Dec. 2017

- Co-founded XOIA, a consulting company specializing in Augmented Reality (AR), with a team of 16 employees.
- We developed multiple **cross-platform Unity Apps.**, web applications using HTML5, CSS, and JavaScript, and native Android apps.
- Our solutions were applied across key sectors, including tourism, culture, public services, and construction. See Demo 1 and Demo 2.

Education

ETH Zurich

RESEARCH INTERNSHIP IN CVG GROUP (MARC POLLEFEYS)

Zurich, Switzerland

2021

- Trained a **Graph Neural Network** for image matching with points and lines; extended work later published in ICCV.
- Contributed to a camera pose estimation library based on points and lines.
- Participated in the data collection efforts for LaMAR Dataset.

Universidad Politécnica de Madrid

PHD IN ARTIFICIAL INTELLIGENCE (CUM LAUDE)

Madrid, Spain

2016 – 2021

- Industrial PhD focused on XR for urban environments.
- Managed a team of 3 students working in synthetic data generation with CARLA simulator, vanishing point estimation, and feature description.
- Teaching assistant for Pattern Recognition and Deep Learning.

Universidad Politécnica de Madrid

M.S. IN ARTIFICIAL INTELLIGENCE

Madrid, Spain

2015 – 2016

- Master Thesis: Prototype of a 3D Augmented Reality Engine for Mobile Devices.
- Key Subjects: Computer Vision, Machine Learning, Bayesian Networks, Evolutionary computation.

Universidade da Coruña

B.S. COMPUTER SCIENCE

A Coruña, Spain

2010 – 2015

Selected Publications

- **ICCV-W 2025**, LightGlueStick: a Fast and Robust Glue for Joint Point-Line Matching.
- **ICCV 2023**, GlueStick: Robust Image Matching by Sticking Points and Lines Together.
- **PR 2022**: ELSEd: Enhanced Line Segment Drawing..
- **RA-L 2021**: Revisiting Binary Local Image Description for Resource Limited Devices.
- **IROS 2018**: FSG: A statistical approach to line detection via fast segments grouping.

Honors & Awards

- 2024 **Excellent PhD Award**, Computer Science Faculty, Universidad Politécnica de Madrid
- 2022 **Reading Group Competition Award**, International Computer Vision Summer School (ICVSS)
- 2019 **Best Student Paper Award**, ibPRIA 2019
- 2019 **Best PhD Thesis Award (ICT field)**, Polytechnic University of Madrid
- 2015 **Best Bachelor's Thesis Award (Open-Source Software)**, Universidade da Coruña